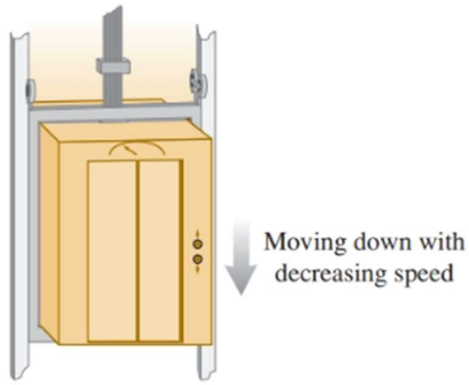
Newton's Laws of Motion Revision

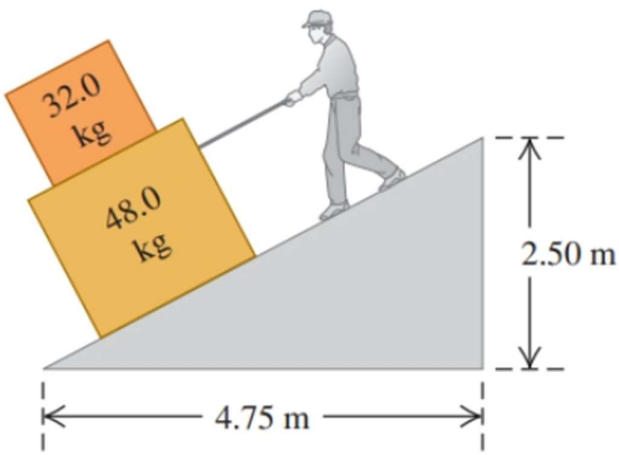
1. An iceboat is on a horizontal, frictionless surface as shown below. It is initially stationary. A wind is blowing from left to right, causing the velocity of the iceboat to increase to 6.0 m s^{-1} over 4.0 seconds. Calculate the horizontal force that the wind exerts on the iceboat. The combined mass of the iceboat and rider is 200.0 kg.

2. An elevator and its load have a combined mass of 800.0 kg as shown below. The elevator is initially moving downwards at 10.0 m s^{-1}. It slows to a stop with constant deceleration over a distance of 25.0 m. What is the tension in the supporting cable while the elevator is slowing down?

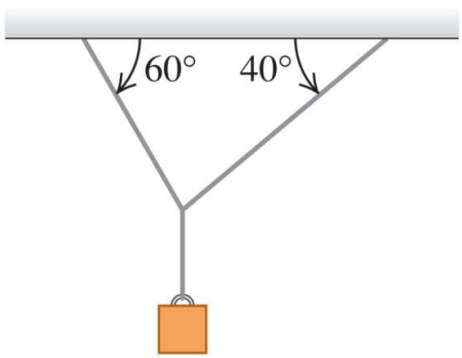


3. A box with mass 11.2 kg is pushed on a horizontal surface with a constant speed of 3.50 m s^{-1}. The force of friction between the box and the surface is 21.952 N.
   a. What horizontal force must be applied to maintain the motion?
   b. If the force calculated in part a is removed, how far does the box slide before coming to rest?

4. Two boxes, one on top of the other, are being lowered down a ramp as shown below. The frictional force between the boxes and the ramp is 308.04 N. Calculate the force required for the boxes to move down the ramp at a constant speed of 0.150 m s^{-1}.



5. Two ropes are connected to a steel cable that supports a hanging weight as shown.



   a. Draw a freebody diagram showing all of the forces acting at the knot that connects the two ropes to the steel cable.
   b. Which of the two ropes will have the greater tension?
   c. If the maximum tension each rope can hold without breaking is 5000.0 N, determine the maximum value of the hanging weight that these ropes can support. Ignore the weight of the ropes and the steel cable.

Newton's Laws of Motion Revision – SOLUTIONS

1. An iceboat is on a horizontal, frictionless surface as shown below. It is initially stationary. A wind is blowing from left to right, causing the velocity of the iceboat to increase to 6.0 m s^{-1} over 4.0 seconds. Calculate the horizontal force that the wind exerts on the iceboat. The combined mass of the iceboat and rider is 200.0 kg.

$$\mathbf{a = \frac{v - u}{t}}$$

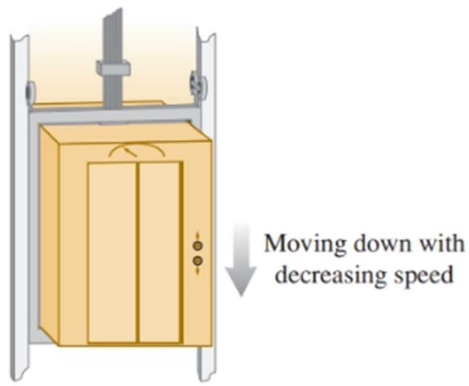
$$\mathbf{a = \frac{6.0 - 0}{4.0} = 1.5\ m\ s^{-2}}$$

F = *ma*

F = 200.0 × 1.5

F = 300 N

2. An elevator and its load have a combined mass of 800.0 kg as shown below. The elevator is initially moving downwards at 10.0 m s^{-1}. It slows to a stop with constant deceleration over a distance of 25.0 m. What is the tension in the supporting cable while the elevator is slowing down?



$v^2 = u^2 + 2as$

$0^2 = 10.0^2 + 2 \times a \times 25.0$

$0 = 100 + 50a$

$a = -2.00 \text{ m s}^{-2}$

$\Sigma F = T - F_W$

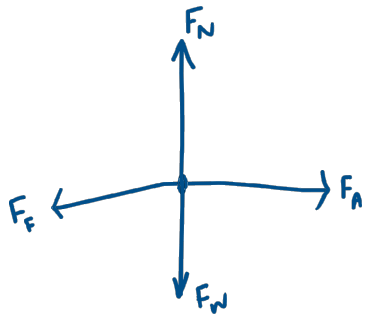
$ma = T - mg$

$800.0 \times (-2.00) = T - 800.0 \times 9.8$

$-1600 = T - 7840$

$T = 6240 \text{ N}$

3. A box with mass 11.2 kg is pushed on a horizontal surface with a constant speed of 3.50 m s $^{-1}$. The force of friction between the box and the surface is 21.952 N.
   a. What horizontal force must be applied to maintain the motion?

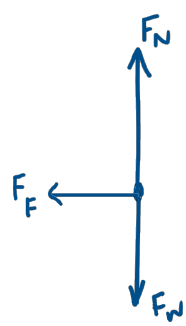


Constant velocity → net force is zero

$F_{Applied} = F_{Friction} = 21.952$ N

The applied force must be in the opposite direction to the friction.

   b. If the force calculated in part a is removed, how far does the box slide before coming to rest?



$\Sigma F = -F_F = -21.952$ N

$$a = \frac{F}{m} = \frac{-21.952}{11.2} = -1.96 \text{ m s}^{-2}$$

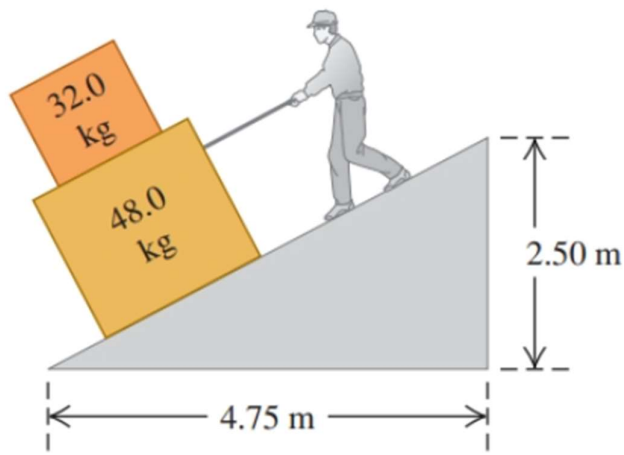
$v^2 = u^2 + 2as$

$0^2 = 3.50^2 + 2 \times (-1.96)s$

$0 = 12.25 - 3.92s$

$s = 3.13$ m

4. Two boxes, one on top of the other, are being lowered down a ramp as shown below. The frictional force between the boxes and the ramp is 308.04 N. Calculate the force required for the boxes to move down the ramp at a constant speed of 0.150 m s^{-1}.



$F_{Wy} = F_W \cos \theta$

$= mg \cos \theta$

$= 80 \times 9.8 \times \cos 28°$

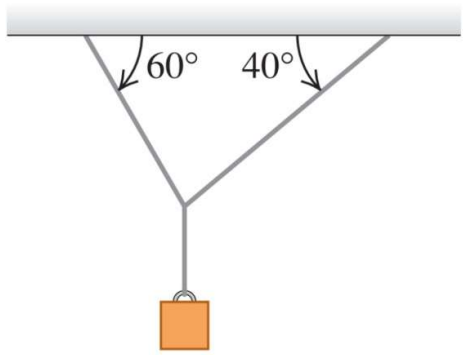
$= 694$ N

$F_{Wx} = T + F_F$ [T is the tension in the rope]

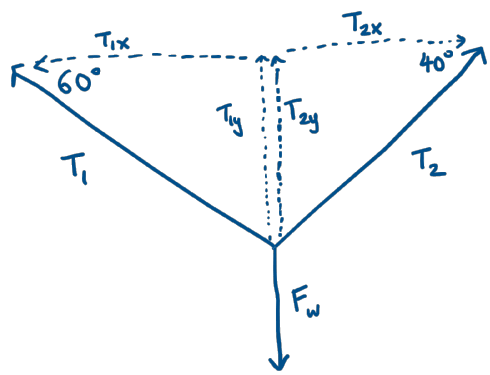
$80 \times 9.8 \times \sin 28° = T + 308.04$

$T = 57.1$ N

5. Two ropes are connected to a steel cable that supports a hanging weight as shown.



a. Draw a freebody diagram showing all of the forces acting at the knot that connects the two ropes to the steel cable.



b. Which of the two ropes will have the greater tension?

$T_{1x} = T_{2x}$

$T_1 \cos 60° = T_2 \cos 40°$

$T_1 \approx 1.53\ T_2$, so $T_1 > T_2$ (the rope that makes a 60° angle with the ceiling has the greater tension)

c. If the maximum tension each rope can hold without breaking is 5000.0 N, determine the maximum value of the hanging weight that these ropes can support. Ignore the weight of the ropes and the steel cable.

$T_1 = 5000$ N

$T_2 = 5000 \times \cos 60° \div \cos 40°$

$T_2 = 3264$ N

$F_W = T_1 \sin 60° + T_2 \sin 40°$

$= 5000 \sin 60° + 3264 \sin 40°$

$= 6428$ N

$= 6400$ N (2 s.f.)